Highlights

Dynamic ICU Bed Allocation Under Resource Scarcity: A Discrete-Event Simulation Integrating Multi-Type Beds and Triage Policies

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- Dynamic simulation of four ICU bed pools grounded in CNES capacities and Epimed targets.
- Separates measured registry metrics, calibrated parameters, and counterfactual assumptions.
- Benchmarks online policies against hindsight and duration-expectation LP upper bounds.
- CMU-MGS saves 18.7 more expected lives than FCFS at $\rho = 1.6$ (4.8% gap to $\mathbb{E}$[LOS] LP bound).
- Evaluates reneging mortality payoffs, negative β coupling, and adult–coronary pooling.

Dynamic ICU Bed Allocation Under Resource Scarcity: A Discrete-Event Simulation Integrating Multi-Type Beds and Triage Policies

Carlos Eduardo Veras Gomes[a], João Augusto Ferreira Nunes de Carvalho[a], Francisco Glaubos Nunes Clímaco[a,*]

[a]Federal University of Maranhão (UFMA), Av. dos Portugueses, 1966, Bacanga, São Luís, 65080-805, MA, Brazil

Abstract

During acute public health crises, epidemic surges, and mass-casualty incidents, intensive care unit (ICU) demand frequently exceeds physical bed availability. Conventional optimization models often formulate ICU triage as a static, single-batch portfolio selection problem with homogeneous beds, overlooking continuous stochastic arrivals, waiting-time clinical deterioration, severity-correlated length of stay (LOS), and specialty barriers across critical care units. This paper presents a continuous-time discrete-event simulation (DES) framework and retrospective time-indexed linear programming (LP) upper bounds for dynamic multi-type ICU bed allocation. The model combines empirical bed capacities from the Brazilian National Register of Healthcare Establishments (DATASUS/CNES) for the University Hospital of the Federal University of Maranhão (HU-UFMA; 55 critical care beds across adult, coronary, pediatric, and neonatal units) with specialty-specific LOS and hospital mortality targets from the Brazilian ICU Registry (AMIB/Epimed), while explicitly separating measured registry statistics, calibrated survival parameters, and counterfactual assumptions. We evaluate six online admission rules—First-Come, First-Served (FCFS), Most Serious First (MSF), Least Serious First (LSF), Marginal Gain in Survival (MGS), a benefit-per-bed-day index (CMU-MGS), and a heuristic deadline-urgency extension (AUSTERE)—alongside two policy-independent offline LP upper bounds across 100 Common Random Numbers (CRN) replications over a 60-day steady-state horizon. Under severe congestion ($\rho = 1.60$), CMU-MGS and MGS save 18.66 (95% CI [17.85, 19.48], +3.64%, $p_{\text{Holm}} < 0.001$) and 17.19 (+3.35%) more expected lives than FCFS, respectively, while cutting mean waiting times of admitted patients from 18.27 to 8.99 hours. Against a hindsight LP bound that knows realized individual stays (LOS_i) and cherry-picks ultra-short realizations, the pessimistic gap of CMU-MGS at $\rho = 1.60$ is 9.89%; when evaluated against a tighter duration-expectation LP bound using $\mathbb{E}[\text{LOS}_i]$ (matching the online information set on stay durations), the gap drops to 4.84%. A 24-cell sensitivity analysis across severity–LOS coupling ($\beta \in [-0.6, 2.4]$) and queue deterioration ($\alpha_{\text{det}} \in [0.05, 0.50]$ day^{-1}) shows that CMU-MGS saves up to +4.99 additional lives over greedy MGS ($p_{\text{Holm}} < 0.001$) when severe patients require longer stays ($\beta \geq 1.2$), and remains robust when reneging entails partial or total pre-ICU mortality ($\theta_{\text{ren}} \in \{0.5, 0.0\}$) as well as under simultaneous clinical estimation noise ($\sigma_{\text{noise}} \leq 0.20$).

Keywords: Healthcare Operations Research, Intensive Care Units, Discrete-Event Simulation, Triage Policies, Queueing with Abandonment, Linear Programming Bounds

1. Introduction

Intensive Care Units (ICUs) constitute a high-acuity, strictly capacity-constrained resource within tertiary healthcare systems. During epidemic waves, seasonal respiratory surges, or mass-casualty disasters, the arrival rate of critically ill patients requiring invasive organ support can rapidly exceed available bed capacity [1, 2]. Under acute scarcity, clinical leadership must rely on transparent triage protocols to determine which candidates are admitted to intensive care and which remain in general or transitional wards [3, 4, 5].

In day-to-day hospital routines, beds are predominantly assigned on a First-Come, First-Served (FCFS) basis. While simple to administer, chronological queueing ignores heterogeneity in baseline mortality risk, therapeutic benefit from intensive care, rate of physiological deterioration while waiting, and expected bed occupation duration [6, 7]. To support rational decision-making, recent studies have proposed portfolio optimization and multi-attribute utility models for ICU bed allocation [6, 8]. Although such models formalize prognostic scoring, they typically treat triage as a static, one-shot batch assignment in a single homogeneous bed pool.

In hospital operations, however, critical care allocation is governed by four interacting dynamic mechanisms:

(i) Asynchronous Arrivals and Bed Turnover: Patients arrive continuously over time, while bed vacan-

*Corresponding author

Email addresses: carlos.evg@discente.ufma.br (Carlos Eduardo Veras Gomes), augusto.joa@discente.ufma.br (João Augusto Ferreira Nunes de Carvalho), francisco.glaubos@ufma.br (Francisco Glaubos Nunes Clímaco)

cies arise stochastically as admitted patients recover, transfer to step-down wards, or die [9, 10, 11].

(ii) Waiting-Queue Clinical Deterioration and Abandonment: Candidates held in emergency departments or pre-ICU wards experience progressive physiological instability; prolonged boarding reduces the survival benefit of subsequent ICU admission and can culminate in pre-ICU mortality or clinical reneging [12, 7].

(iii) Severity–Duration Coupling: Critically ill patients who survive frequently require prolonged mechanical ventilation and vasoactive support, inducing a positive correlation between baseline clinical severity and ICU length of stay (LOS) among survivors, whereas ultra-severe patients who suffer early in-ICU death may experience truncated stays [? 13]. When severe patients require longer stays, admitting a candidate with high absolute benefit locks a bed for an extended period, blocking multiple shorter-stay candidates.

(iv) Multi-Type Specialization and Partial Pooling: Tertiary hospitals partition critical care into specialized units—neonatal, pediatric, general adult, and coronary care—due to equipment and clinical skill barriers [14, 15]. While pediatric and neonatal pools are strictly segregated, adult medical and coronary care units (UCO) can partially share compatible adult patients during surge conditions [16].

To capture these operational realities, this paper develops a continuous-time discrete-event simulation (DES) framework coupled with retrospective linear programming (LP) upper bounds for multi-type ICU bed allocation. Our main contributions are:

- Dynamic Queueing Formulation with Deterioration and Severity-Coupled LOS: We model a multi-pool $M/G/c + G$ critical care system where post-admission survival probability decays exponentially with waiting time and individual expected LOS scales with baseline severity, creating an explicit operational tension between marginal survival gain and bed-day consumption.

- Transparent Multi-Pool Calibration: Physical bed capacities are taken directly from the Brazilian National Register of Healthcare Establishments (DATASUS/CNES) for the University Hospital of the Federal University of Maranhão (HU-UFMA: 55 beds across adult, coronary, pediatric, and neonatal units) [17], while specialty-specific LOS distributions and post-admission mortality rates are calibrated to match national Brazilian ICU registries (AMIB/Epimed, ORCHESTRA, and the Brazilian Neonatal Research Network) [18, 19? , 15, 12], with an explicit separation between measured data, calibrated parameters, and counterfactual assumptions.

- Comprehensive Policy Benchmarking and Tightened LP Upper Bounds: Using a 100-replication Common Random Numbers (CRN) design with steady-state warm-up and complete post-horizon cohort drainage, we compare six online rules—FCFS, Most Serious First (MSF), Least Serious First (LSF), Marginal Gain in Survival (MGS), a $c\mu$-type survival-rate index (CMU-MGS) [20], and a heuristic deadline-urgency extension (AUSTERE)—against both a clairvoyant hindsight LP bound (using realized LOS_i) and a tighter duration-expectation LP bound (using $\mathbb{E}[\mathrm{LOS}_i]$ in the spirit of information relaxations [21]).

- Parametric Regime Mapping, Reneging Payoff Sensitivity, and Imperfect Information Analysis: Through systematic sensitivity experiments across positive and negative severity–LOS coupling ($\beta \in [-0.6, 2.4]$), queue deterioration (α_{det}), reneging survival payoffs ($\theta_{\mathrm{ren}} \in \{1.0, 0.5, 0.0\}$), adult–coronary overflow rules, and multi-parameter clinical estimation noise (σ_{noise}) [22], we characterize the operational conditions under which duration-aware index rules outperform greedy triage.

The remainder of this paper is organized as follows. Section 2 reviews the operations research literature on ICU management and dynamic triage. Section 3 formalizes the queueing model, triage indices, and offline LP upper bounds. Section 4 details the empirical parameterization and calibration taxonomy. Section 5 presents the simulation results, offline optimality gaps, sensitivity analyses, and ablation studies. Section 6 discusses clinical, ethical, and methodological limitations, and Section 7 concludes.

2. Related Work

The operations research and management science (OR/MS) literature on intensive care capacity management and patient prioritization spans empirical econometrics, Markov decision processes (MDPs), queueing theory with abandonment, and discrete-event simulation. Table 1 summarizes representative studies and contrasts their modeling assumptions with the present work.

A foundational stream of OR/MS research examines how ICU congestion shapes admission and discharge decisions. Chan et al. [9] formulated an optimization model for demand-driven early ICU discharge, accounting for post-discharge readmission risks and deriving an index policy that prioritizes discharging patients with low marginal readmission cost per bed-day freed. Kim et al. [10] conducted an econometric and simulation study of 15 Kaiser Permanente hospitals, demonstrating that ICU occupancy at arrival significantly alters admission probability and that boarding delays measurably worsen patient outcomes. More recently, Valeva et al. [16] analyzed the interaction between ICUs and downstream step-down units under flexible staffing, deriving dual-based acuity policies to mitigate

Table 1: Comparison of the proposed framework with representative ICU operations and triage models in the literature.

Study	Methodology	Bed Pools	Arrival Process	LOS Model	Queue Decay / Reneging	Triage / Control Policy	Offline Bound
Chan et al. [9]	MDP / Simulation	Single ICU	Stochastic	Geometric / Empirical	No (Readmission risk)	Discharge index policy	Dynamic prog.
Mills et al. [23]	Fluid / Queueing	Multi-resource	Static / Surge	Exp. / Resource-tied	Time-varying survival	Dynamic $c\mu$ / Myopic index	Fluid optimal
Kim et al. [10]	Econometric / Sim.	Single ICU	Empirical Poisson	Log-Normal / Empirical	Delayed admission cost	Threshold admission control	No
Sun et al. [7]	MDP / Queueing	Single server/pool	None / Clearing	Exponential (2 classes)	Discrete deterioration	Austere priority index	MDP optimal
Alban et al. [24]	Process DES	Single ICU	Poisson	Empirical / Weibull	Diversion / Overflow	Capacity planning threshold	No
Bilbao [11]	DES / Flight Sim.	Single ICU + Ward	Gompertz / Poisson	Empirical / Phase-type	Early discharge / Divert	Physician behavioral rules	No
Valeva et al. [16]	Two-stage LP / DES	ICU + Step-down	Markovian	Exponential	Bed blocking	Acuity-based outflow LP	Stationary dual
Frej et al. [8]	Static ILP / Utility	Single ICU	None (Batch)	Not modeled	No	Bayesian SOFA portfolio	Exact static ILP
Murray et al. [25]	Time-Series / Surv.	Single ICU	ARIMA	Cox / Weibull (MODS)	No	Census forecasting	No
This Work	DES + LP Bounds	4 Pools (Coupled)	Continuous Poisson	Severity-Log-Normal	Exp. Decay + Reneging	FCFS, MSF, LSF, MGS, CMU, AUSTERE	Time-indexed LP Bounds

bed blocking, while Bilbao [11] and Murray et al. [25] combined survival analysis with arrival forecasting to predict short-term ICU census.

A second stream investigates casualty and critical care triage under acute resource scarcity, drawing heavily on the $c\mu$ scheduling rule [20] and queueing systems with impatient or deteriorating customers. Argon and Ziya [22] studied priority assignment when customer types are observed imperfectly, showing that classification noise degrades multi-class rules and favors robust index structures. Mills et al. [23] modeled mass-casualty prioritization across multiple bottleneck resources, showing that fluid-based dynamic indices accounting for both survival deterioration and resource processing rates outperform static color-tag triage (such as Simple Triage and Rapid Treatment). Closest to our clinical triage focus, Sun et al. [7] analyzed patient prioritization under austere conditions where queued patients transition between severity states and may abandon (die) before service. They proved that optimal prioritization balances the immediate marginal survival reward against expected service time and the hazard of clinical deterioration.

During the COVID-19 pandemic, several studies addressed hospital surge allocation. Alban et al. [24] built a discrete-event simulation (icu-covid-sim) to project regional overflow probabilities. White and Lo [5] and Aquino et al. [2] evaluated ethical multi-principle frameworks combining short-term prognosis with equity considerations. In the Brazilian context, Frej et al. [6, 8] proposed a binary portfolio optimization model using Expected Utility Theory and SOFA scores to select which candidates occupy open beds.

Despite these advances, Table 1 highlights an important methodological gap. Analytical triage models such as Sun et al. [7] and Mills et al. [23] rely on exponential service times and single-pool or clearing formulations to preserve mathematical tractability, whereas utility-based models like Frej et al. [8] omit continuous arrivals, bed turnover, and queue waiting decay entirely. Conversely, multi-unit hospital simulations typically apply simple FCFS or threshold rules without benchmarking $c\mu$-type indices or quantifying the optimality gap against retrospective mathematical programming bounds. Our work bridges this gap by embedding $c\mu$ and deadline-urgency indices within a continuous-time, multi-specialty hospital simulation calibrated with Brazilian intensive care registry targets, while evaluating their distance to both hindsight and duration-expectation time-indexed LP upper bounds.

3. Dynamic Simulation and Optimization Modeling

3.1. System Architecture and Queueing Dynamics

We consider a tertiary referral hospital operating over a continuous time horizon $[0, T_{\text{warm}} + T)$, where $T_{\text{warm}} > 0$ is an initial warm-up period used to reach stochastic steady state and $[T_{\text{warm}}, T_{\text{warm}} + T)$ is the active evaluation window of length T days. The critical care department comprises four specialized bed pools $J = \{\text{adult}, \text{coronary}, \text{pediatric}, \text{neonatal}\}$, with physical bed capacities $C_j \in \mathbb{Z}_+$ $(j \in J)$ and total capacity $C_{\text{total}} = \sum_{j \in J} C_j$.

Patients requiring intensive care in specialty $j \in J$ arrive according to an independent homogeneous Poisson process with rate $\lambda_j > 0$ (patients per day). Each arriving patient i is characterized by the attribute vector:

$$\boldsymbol{\xi}_i = \left(j(i), t^i_{\text{arr}}, P^i_{\text{out}}, P^i_{\text{in},0}, \alpha^{j(i)}_{\text{det}}, \mathbb{E}[\text{LOS}_i], \text{LOS}_i, W^i_{\max}, u_i \right) \quad (1)$$

where:

- $j(i) \in J$ is the patient's primary clinical specialty;
- $t^i_{\text{arr}} \geq 0$ is the arrival epoch;
- $P^i_{\text{out}} \in (0, 1)$ is the baseline probability of hospital survival if the patient is never admitted to an ICU bed and is managed in general or step-down wards;
- $P^i_{\text{in},0} \in [P^i_{\text{out}}, 1)$ is the initial probability of survival if admitted immediately upon arrival $(w_i = 0)$, parameterized as $P^i_{\text{in},0} = P^i_{\text{out}} + (1 - P^i_{\text{out}})g_i$, where $g_i \in (0, 1)$ is the fractional therapeutic efficacy of ICU care;
- $\alpha^{j(i)}_{\text{det}} > 0$ is the specialty-specific clinical deterioration rate (day^{-1}) during pre-ICU queue boarding;
- $\mathbb{E}[\text{LOS}_i] > 0$ and $\text{LOS}_i > 0$ are the expected and realized ICU lengths of stay (in days) once admitted;
- $W^i_{\max} > 0$ is the finite queue waiting tolerance (deadline in days) before clinical reneging (which in hospital operations may represent emergency transfer, clinical stabilization/withdrawal of ICU request in a ward, or pre-ICU death);
- $u_i \sim \text{Uniform}(0, 1)$ is an independent uniform random variate used to realize binary survival outcomes under Common Random Numbers.

3.1.1. Waiting-Time Clinical Deterioration and Reneging Payoff

Clinical studies consistently show that each hour of delayed ICU transfer increases mortality risk due to progression of shock and organ dysfunction [12, 10]. If patient i waits $w = t - t^i_{\text{arr}} \in [0, W^i_{\max}]$ days in the pre-ICU queue before admission at time t, the post-admission survival probability decays exponentially toward the baseline ward survival P^i_{out}:

$$P^i_{\text{in}}(w) = P^i_{\text{out}} + \left(P^i_{\text{in},0} - P^i_{\text{out}}\right)\exp\left(-\alpha^{j(i)}_{\text{det}} w\right) \tag{2}$$

If patient i is not admitted within $W^i_{\max}$ days, the patient reneges from the queue at epoch $t^i_{\text{arr}} + W^i_{\max}$. To account for the fact that queue abandonment may represent a mixture of ward management/transfer (survival P^i_{out}) and pre-ICU mortality (survival 0), we parameterize the expected survival payoff of a reneging patient as $\theta_{\text{ren}} P^i_{\text{out}}$, where $\theta_{\text{ren}} \in [0, 1]$ is the reneging survival retention factor. In the baseline model ($\theta_{\text{ren}} = 1.0$), reneging patients receive the ward survival probability P^i_{out}, and the instantaneous marginal survival gain of admitting patient i after waiting time $w = t - t^i_{\text{arr}}$ is:

$$\Delta P_i(w) = P^i_{\text{in}}(w) - P^i_{\text{out}} = \left(P^i_{\text{in},0} - P^i_{\text{out}}\right)\exp\left(-\alpha^{j(i)}_{\text{det}} w\right) \tag{3}$$

When $\theta_{\text{ren}} < 1.0$ (analyzed in Section 5.4.3), avoiding a reneging event yields the larger net survival increment $P^i_{\text{in}}(w) - \theta_{\text{ren}} P^i_{\text{out}} = \Delta P_i(w) + (1 - \theta_{\text{ren}}) P^i_{\text{out}}$.

3.1.2. Severity-Correlated Length of Stay

To reflect the clinical relationship between acute physiological severity and duration of organ support [? 13], the individual expected stay $\mathbb{E}[\text{LOS}_i]$ is coupled to the patient's baseline mortality risk $1 - P^i_{\text{out}}$:

$$\mathbb{E}[\text{LOS}_i] = \mu_{\text{los},j(i)} \cdot \max\left\{0.35,\ 1 + \beta\left((1 - P^i_{\text{out}}) - \mathbb{E}[1 - P^{j(i)}_{\text{out}}]\right)\right\} \tag{4}$$

where $\mu_{\text{los},j}$ is the empirical mean LOS of specialty j, $\mathbb{E}[1 - P^j_{\text{out}}]$ is the mean baseline severity in specialty j, and $\beta \in \mathbb{R}$ governs the direction and strength of the severity–duration coupling ($\beta > 0$ when higher severity prolongs stay among survivors; $\beta < 0$ when extreme severity leads to shorter stays due to early in-ICU mortality). Because P^i_{out} spans a width of 0.40 in every pool (Table 2), the centered deviation $(1 - P^i_{\text{out}}) - \mathbb{E}[1 - P^j_{\text{out}}]$ lies strictly in $[-0.20, +0.20]$. Consequently, for all $|\beta| \le 2.4$ considered in this study, the linear factor $1 + \beta((1 - P^i_{\text{out}}) - \mathbb{E}[1 - P^j_{\text{out}}])$ is bounded below by $1 - 2.4(0.20) = 0.52 > 0.35$. Thus, the safety floor of 0.35 is never active for $|\beta| \le 2.4$, guaranteeing mathematically that the unconditional mean stay across arriving patients in pool j is exact: $\mathbb{E}[\text{LOS}_i] = \mu_{\text{los},j}$.

Given $\mathbb{E}[\text{LOS}_i]$ and the specialty's coefficient of variation $CV_{\text{los},j}$, the realized stay LOS_i follows a Log-Normal distribution $\text{LogNormal}(\mu_i, \sigma_j^2)$ with exact log-scale parameters:

$$\sigma_j = \sqrt{\ln\left(1 + CV^2_{\text{los},j}\right)}, \qquad \mu_i = \ln(\mathbb{E}[\text{LOS}_i]) - \frac{1}{2}\sigma_j^2 \tag{5}$$

3.1.3. Hierarchical Multi-Pool Compatibility and Overflow

Pediatric (j = pediatric) and neonatal (j = neonatal) units require dedicated ventilators, incubators, and subspecialty nursing, operating as strictly segregated pools. Conversely, general adult ICU beds (j = adult) and adult coronary care beds (j = coronary) share adult life-support infrastructure and can operate under a hierarchical partial-pooling (overflow) protocol:

(1) Primary Home-Pool Priority: When a bed in pool $j \in \{\text{adult}, \text{coronary}\}$ becomes vacant at time t, if the native queue $Q_j(t) \neq \emptyset$, the bed is assigned exclusively to the highest-priority candidate in $Q_j(t)$ according to active triage policy π.

(2) Secondary Cross-Pool Overflow: If the native queue is empty ($Q_j(t) = \emptyset$) and the coupled adult pool $j' \in \{\text{adult}, \text{coronary}\} \setminus \{j\}$ has waiting candidates ($Q_{j'}(t) \neq \emptyset$), the vacant bed in j admits the highest-priority patient from $Q_{j'}(t)$ under policy π. Similarly, an arriving adult or coronary patient finding their native unit full is immediately admitted to the partner unit if an idle bed is available there.

Because cross-pool admission occurs only when the home queue is empty, candidates from different specialty pools never compete simultaneously for the same bed vacancy; within any non-empty queue, candidates belong to the same specialty and are ordered strictly by policy score $s^\pi_i(t)$.

3.2. Online Triage Policies

Whenever a bed vacancy occurs at epoch t in an eligible queue $Q(t) \neq \emptyset$, triage policy π admits patient i^*:

$$i^* = \arg\max_{i \in Q(t)} \left\{s^\pi_i(t)\right\}, \quad \text{with ties broken by earliest } t^i_{\text{arr}} \tag{6}$$

We evaluate six online prioritization policies:

1. First-Come, First-Served (FCFS): Prioritizes candidates in chronological order of arrival:

$$s^{\text{FCFS}}_i(t) = -t^i_{\text{arr}} \tag{7}$$

2. Most Serious First (MSF): Prioritizes the sickest candidates with the highest baseline mortality risk without ICU care:

$$s^{\text{MSF}}_i(t) = 1 - P^i_{\text{out}} \tag{8}$$

MSF prioritizes acute severity regardless of whether the patient has high therapeutic responsiveness or prolonged expected stay.

3. Least Serious First (LSF): Prioritizes candidates with the highest absolute post-ICU survival probability at decision time t:

$$s^{\text{LSF}}_i(t) = P^i_{\text{in}}(t - t^i_{\text{arr}}) \tag{9}$$

4. Marginal Gain in Survival (MGS): Prioritizes candidates with the largest instantaneous incremental survival benefit from ICU admission (incorporating any additional reneging survival loss when $\theta_{\text{ren}} < 1$):

$$s_i^{\text{MGS}}(t) = P_{\text{in}}^i(t - t_{\text{arr}}^i) - \theta_{\text{ren}} P_{\text{out}}^i \tag{10}$$

5. Benefit-per-Bed-Day Index (CMU-MGS): Motivated by the classical $c\mu$ scheduling rule [20, 23], CMU-MGS normalizes the instantaneous marginal survival gain by the patient's expected bed-days $\mathbb{E}[\text{LOS}_i]$:

$$s_i^{\text{CMU-MGS}}(t) = \frac{P_{\text{in}}^i(t - t_{\text{arr}}^i) - \theta_{\text{ren}} P_{\text{out}}^i}{\max(0.2,\ \mathbb{E}[\text{LOS}_i])} \tag{11}$$

CMU-MGS maximizes the rate of expected lives saved per unit of bottleneck bed capacity consumed.

6. CMU-MGS with Heuristic Deadline Urgency (AUSTERE): Motivated qualitatively by the observation in austere-conditions triage [7] that queued patients face an approaching abandonment deadline, we also evaluate a heuristic deadline-urgency extension of CMU-MGS. Because candidates from different pools never compete against one another (Section 3.1.3), any pool-constant deterioration parameter $\alpha_{\text{det}}^{j(i)}$ is invariant across competing candidates in $Q_j(t)$ and is therefore omitted. Let $\tau_i(t) = \max\left(\tau_{\min},\ t_{\text{arr}}^i + W_{\max}^i - t\right)$ denote the floored remaining time to patient i's reneging deadline. The heuristic score multiplies CMU-MGS by a bounded deadline-urgency factor:

$$s_i^{\text{AUST}}(t) = s_i^{\text{CMU-MGS}}(t)\left(1 + \frac{\kappa}{\tau_i(t) + \kappa}\right) \tag{12}$$

with baseline smoothing constants $\tau_{\min} = 0.25$ days and $\kappa = 0.5$ days (so that the urgency weight increases smoothly from ≈ 1.17 for newly arrived patients to 1.67 as the deadline approaches). We emphasize that Eq. (12) is a heuristic benchmark rather than an MDP-optimal index; shifting $(\tau_{\min}, \kappa)$ merely interpolates between pure CMU-MGS ($\kappa \to 0$) and aggressive deadline-first ordering ($\kappa \gg 1$, which degrades survival by admitting long-stay candidates solely because their tolerance $W_{\max}^i$ is expiring).

3.2.1. Triage Under Imperfect Clinical Information

In clinical practice, prognostic probabilities and expected stay durations are estimated from physiological scores (e.g., SAPS 3, SOFA, PIM 3, SNAPPE-II) subject to estimation error [22, 8]. To evaluate informational robustness fairly across all clinical policies, when observation noise $\sigma_{\text{noise}} > 0$ is present, the triage controller observes noisy estimates $(\widetilde{P}_{\text{out}}^i, \widetilde{P}_{\text{in},0}^i, \widetilde{\mathbb{E}}[\text{LOS}_i])$ for every patient:

$$\begin{aligned} \widetilde{P}_{\text{out}}^i &= \text{clip}_{[0.01,0.95]}\left(P_{\text{out}}^i + \varepsilon_{i,1}\right), \\ \widetilde{P}_{\text{in},0}^i &= \text{clip}_{[\widetilde{P}_{\text{out}}^i, 0.99]}\left(P_{\text{in},0}^i + \varepsilon_{i,2}\right), \\ \widetilde{\mathbb{E}}[\text{LOS}_i] &= \mathbb{E}[\text{LOS}_i] \cdot \max(0.4,\ 1 + \varepsilon_{i,3}) \end{aligned} \tag{13}$$

where $\varepsilon_{i,1}, \varepsilon_{i,2}, \varepsilon_{i,3} \sim \mathcal{N}(0, \sigma_{\text{noise}}^2)$ are independent Gaussian perturbations. All non-FCFS policies (MSF, LSF, MGS, CMU-MGS, and AUSTERE) compute their priority scores using these noisy estimates, whereas actual post-triage survival outcomes are evaluated using the true underlying parameters $(P_{\text{out}}^i, P_{\text{in},0}^i)$.

3.3. Evaluation Cohort, Post-Horizon Drainage, and Metrics

To eliminate initialization bias and boundary truncation distortions, each simulation replication generates arrivals over $[0, T_{\text{warm}} + T)$ and continues executing discharge, admission, and timeout events beyond $T_{\text{warm}} + T$ until every patient who arrived during the evaluation window $[T_{\text{warm}}, T_{\text{warm}} + T)$ has either been admitted to a bed or reneged from the queue. Let $\mathcal{I}_{\text{eval}} = \{i : T_{\text{warm}} \le t_{\text{arr}}^i < T_{\text{warm}} + T\}$ denote the evaluated patient cohort of size $N_{\text{eval}} = |\mathcal{I}_{\text{eval}}|$, partitioned into the admitted subset $\mathcal{A}^\pi \subseteq \mathcal{I}_{\text{eval}}$ and its complement (the reneged/unadmitted subset) $\mathcal{R}^\pi = \mathcal{I}_{\text{eval}} \setminus \mathcal{A}^\pi$. Because simulation continues until all queues drain, $\mathcal{A}^\pi \cup \mathcal{R}^\pi = \mathcal{I}_{\text{eval}}$ holds exactly.

System performance over $\mathcal{I}_{\text{eval}}$ is quantified by:

- Expected Survivors:

$$S_{\text{exp}}^\pi = \sum_{i \in \mathcal{A}^\pi} P_{\text{in}}^i\left(t_{\text{adm}}^i - t_{\text{arr}}^i\right) + \sum_{i \in \mathcal{R}^\pi} \theta_{\text{ren}} P_{\text{out}}^i \tag{14}$$

- Realized Survivors:

$$S_{\text{real}}^\pi = \sum_{i \in \mathcal{A}^\pi} \mathbb{I}\left(u_i < P_{\text{in}}^i(t_{\text{adm}}^i - t_{\text{arr}}^i)\right) + \sum_{i \in \mathcal{R}^\pi} \mathbb{I}\left(u_i < \theta_{\text{ren}} P_{\text{out}}^i\right) \tag{15}$$

- In-Hospital Mortality Rate of Admitted Patients: To verify internal clinical consistency against ICU registry mortality reports (which measure mortality exclusively among admitted ICU patients), we record the expected mortality rate among admitted patients in pool j:

$$M_{\text{adm},j}^\pi = \frac{1}{|\mathcal{A}_j^\pi|} \sum_{i \in \mathcal{A}_j^\pi} \left(1 - P_{\text{in}}^i(t_{\text{adm}}^i - t_{\text{arr}}^i)\right) \tag{16}$$

- Disaggregated Waiting Times and Reneging Rate: We report the reneging rate $r_{\text{ren}}^\pi = |\mathcal{R}^\pi|/N_{\text{eval}}$, the mean waiting time of admitted patients $\bar{W}_{\text{adm}}^\pi = \frac{24}{|\mathcal{A}^\pi|} \sum_{i \in \mathcal{A}^\pi} (t_{\text{adm}}^i - t_{\text{arr}}^i)$ (in hours), and the mean waiting time prior to abandonment for reneged patients $\bar{W}_{\text{ren}}^\pi = \frac{24}{|\mathcal{R}^\pi|} \sum_{i \in \mathcal{R}^\pi} W_{\max}^i$.

- Steady-State Time-Weighted Occupancy: Measured strictly over the steady-state observation window $[T_{\text{warm}}, T_{\text{warm}} + T)$:

$$\bar{O}_j^\pi = \frac{1}{T \cdot C_j} \int_{T_{\text{warm}}}^{T_{\text{warm}}+T} B_j^\pi(t)\, dt \tag{17}$$

where $B_j^\pi(t) \in \{0, 1, \ldots, C_j\}$ is the number of occupied beds in pool j at time t.

3.4. Policy-Independent Time-Indexed LP Upper Bounds

To benchmark online policies against theoretical upper bounds without tying the benchmark to the trajectory of any single rule, we formulate two policy-independent time-indexed linear programming (LP) relaxations for each replication. We clarify three structural properties that guarantee the validity of these bounds across all online policies:

(a) Policy-Independent Initial State at T_{warm}: Different online policies π arrive at the end of the warm-up period T_{warm} with different sets of legacy patients occupying beds. If an offline program enforced the legacy bed occupation of a specific policy (e.g., FCFS), it would not be a valid bound for policies that arrived at T_{warm} with fewer occupied beds. Therefore, the LP optimizes strictly over the evaluated cohort $\mathcal{I}_{\text{eval}}$ starting from an empty bed state at $t = T_{\text{warm}}$ (zero legacy occupation from $[0, T_{\text{warm}})$). Because every online policy π faces non-negative bed occupation inherited from the warm-up period on $[T_{\text{warm}}, T_{\text{warm}} + T)$, dropping warm-up occupancy strictly enlarges the feasible capacity region for $\mathcal{I}_{\text{eval}}$, yielding a policy-independent relaxation valid for every π.

(b) Full Adult–Coronary Pooling and Complete Drainage Horizon: When adult–coronary overflow is enabled, the LP relaxes the hierarchical home-pool priority into a single combined group $g_{\text{adult+coronary}}$ with capacity $C_{\text{adult+coronary}} = 15 + 10 = 25$ beds, alongside $C_{\text{pediatric}} = 10$ and $C_{\text{neonatal}} = 20$. Time is discretized into slots $s \in \mathcal{S} = \{0, 1, \ldots, S_{\max}\}$ of length $\Delta t = 0.5$ days starting at T_{warm}, where $S_{\max} = \max_{i \in \mathcal{I}_{\text{eval}}}\{s^i_{\text{arr}} + K_i + L_i\} + 2$ is set dynamically so that $\mathcal{S}$ covers 100% of post-horizon queue drainage and patient stays.

(c) Hindsight LP Bound vs. Duration-Expectation LP Bound: A hindsight LP that uses realized stays LOS_i enjoys perfect clairvoyance over random log-normal stay realizations, allowing it to selectively admit patients who happen to draw ultra-short realized stays ($\text{LOS}_i \ll \mathbb{E}[\text{LOS}_i]$). In the spirit of information relaxations for stochastic dynamic programs [21], we decouple arrival-stream knowledge from service-duration clairvoyance by solving two LP bounds:

- Hindsight LP Bound ($S^{\text{real}}_{\text{UB}}$): Uses each patient's realized stay $L^{\text{real}}_i = \max(1, \lfloor \text{LOS}_i/\Delta t \rfloor)$ slots.
- Duration-Expectation LP Bound ($S^{\text{exp}}_{\text{UB}}$): Uses each patient's expected stay $L^{\text{exp}}_i = \max(1, \lfloor \mathbb{E}[\text{LOS}_i]/\Delta t \rfloor)$ slots, restricting the offline planner to the exact same duration information $\mathbb{E}[\text{LOS}_i]$ available to online policies at triage.

For a given stay-slot vector $L_i \in \{L^{\text{real}}_i, L^{\text{exp}}_i\}$, arrival slot $s^i_{\text{arr}} = \lfloor (t^i_{\text{arr}} - T_{\text{warm}})/\Delta t \rfloor$, maximum wait $K_i = \lceil W^i_{\max}/\Delta t \rceil$, and fractional admission variable $x_{i,k} \in [0, 1]$ at wait slot $k \in \{0, 1, \ldots, K_i\}$, the LP upper bound solves:

$$\max_{x} \quad \sum_{i \in \mathcal{I}_{\text{eval}}} \theta_{\text{ren}} P^i_{\text{out}} + \sum_{i \in \mathcal{I}_{\text{eval}}} \sum_{k=0}^{K_i} \Big[\Delta P_i(\min(W^i_{\max}, k\Delta t)) + (1 - \theta_{\text{ren}}) P^i_{\text{out}} \Big] x_{i,k} \tag{18}$$

$$\text{s.t.} \quad \sum_{k=0}^{K_i} x_{i,k} \le 1, \qquad \forall i \in \mathcal{I}_{\text{eval}} \tag{19}$$

$$\sum_{i \in \mathcal{I}_g} \sum_{\substack{k=0 \\ s - L_i < s^i_{\text{arr}} + k \le s}}^{K_i} x_{i,k} \le C_g, \qquad \forall g, \ \forall s \in \mathcal{S} \tag{20}$$

$$x_{i,k} \in [0, 1], \qquad \forall i \in \mathcal{I}_{\text{eval}}, \ \forall k \in \{0, \ldots, K_i\} \tag{21}$$

where $\Delta P_i(\min(W^i_{\max}, k\Delta t))$ denotes the marginal survival gain function of Eq. (3) evaluated at waiting duration $w = \min(W^i_{\max}, k\Delta t)$. We solve the continuous LP relaxation (18)–(21) directly via the HiGHS solver.

4. Experimental Design and Calibration Taxonomy

4.1. Measured Registry Data vs. Calibrated and Counterfactual Parameters

A central methodological requirement in healthcare simulation is distinguishing quantities directly measured in hospital registries from parameters calibrated to reproduce aggregate clinical targets and counterfactual quantities that cannot be observed in admitted-only databases. Table 2 presents the multi-pool parameterization for HU-UFMA, explicitly classifying every parameter into three epistemological categories: (I) Measured, (II) Calibrated/Derived, and (III) Assumed (Counterfactual/Sensitivity).

(I) Directly Measured Quantities: From the Brazilian National Register of Healthcare Establishments (DATASUS/CNES, Establishment ID 2111302726653, competency 12/2024) [17], HU-UFMA operates $C_{\text{total}} = 55$ critical care beds dedicated to the Unified Health System (SUS): 15 Adult Type III ICU beds, 10 Type III Coronary Care (UCO) beds, 10 Pediatric Type III ICU beds, and 20 Neonatal Type II ICU beds. Specialty-level mean lengths of stay and hospital mortality targets for admitted patients are taken from the AMIB/Epimed UTIs Brasileiras national registry [18] and the multicenter ORCHESTRA studies [19, 13?]: public adult ICUs average $\mu_{\text{los, adult}} = 6.5$ days with $\approx$ 26%–28% hospital mortality; coronary units average $\mu_{\text{los, coronary}} = 4.8$ days with $\approx$ 14%–17% mortality; and pediatric ICUs in the AMIB/Epimed pediatric panel [18] average $\mu_{\text{los, pediatric}} = 6.0$ days with $\approx$ 5%–8% mortality. For the Type II Neonatal ICU pool, $\mu_{\text{los, neonatal}} = 11.5$ days represents the overall unit average across general neonatal admissions (blending term and moderate preterm neonates staying 5–8 days with the subgroup of very-low-birth-weight preterm

Table 2: Multi-pool parameterization for HU-UFMA (DATASUS/CNES competency 12/2024) and Brazilian ICU registries, separating directly measured registry quantities, calibrated/derived parameters, and counterfactual assumptions, alongside model-generated admitted mortality verification.

		(I) Measured (Registry Data)		(II) Calibrated / Derived to Match Registry			(III) Assumed (Counterfactual)	
Bed Pool (j)	CNES Code / Specialty	Beds (C_j)	Mean LOS ($\mu_{\text{los},j}$, CV_j)	Decay (α^j_{det})	Registry Mort. Target	Model Adm. Mort. ($\rho = 1.0$ / 1.6)	Ward Surv. (P^i_{out})	ICU Efficacy (g_i), $W^i_{\max}$
Adult ICU	76 – UTI Adulto Tipo III	15	6.5 d (0.65)	0.18 d^{-1}	≈ 26%–28% [18, 19]	25.3% / 25.5% (FCFS: 26.1% / 30.2%)	$U(0.15, 0.55)$	$U(0.45, 0.75)$, $U(0.5, 2.5)$ d
Coronary (UCO)	86 – UCO Tipo III	10	4.8 d (0.55)	0.20 d^{-1}	≈ 14%–17% [18, 13]	15.2% / 16.3% (FCFS: 15.8% / 19.1%)	$U(0.30, 0.70)$	$U(0.55, 0.85)$, $U(0.5, 2.0)$ d
Pediatric ICU	79 – UTI Pediátrica Tipo III	10	6.0 d (0.65)	0.15 d^{-1}	≈ 5%–8% [18]	6.8% / 8.0% (FCFS: 7.2% / 9.5%)	$U(0.50, 0.90)$	$U(0.65, 0.90)$, $U(0.5, 2.5)$ d
Neonatal ICU	81 – UTI Neonatal Tipo II	20	11.5 d (0.75)	0.20 d^{-1}	≈ 8%–11% [18, 15]	8.6% / 10.0% (FCFS: 9.0% / 11.8%)	$U(0.45, 0.85)$	$U(0.60, 0.88)$, $U(0.5, 2.0)$ d
Total / System	Estab. CNES 2111302726653	**55**	Sources: DATASUS/CNES [17], AMIB/Epimed UTIs Brasileiras [18], ORCHESTRA [19?], RBPN [15], Cardoso et al. [12].					

infants analyzed by the Brazilian Neonatal Research Network [15], whose stays exceed 30–45 days).

(II) Calibrated and Derived Parameters: Because hospital registries record outcomes only for patients admitted to the ICU, the counterfactual survival without ICU care (P^i_{out}) and the fractional ICU therapeutic efficacy (g_i) are not directly observable. We therefore calibrate the uniform support ranges of P^i_{out} and g_i in each pool so that the resulting post-admission mortality rate $1 - P^i_{\text{in}}(w)$ generated by the simulation matches the empirical AMIB/Epimed mortality targets. Specifically, at $w = 0$, the theoretical mean initial mortality $1 - \mathbb{E}[P_{\text{in},0}] = (1 - \mathbb{E}[P_{\text{out}}])(1 - \mathbb{E}[g])$ equals $(1 - 0.35)(1 - 0.60) = 26.0\%$ for Adult ICU, $(1 - 0.50)(1 - 0.70) = 15.0\%$ for Coronary UCO, $(1 - 0.70)(1 - 0.775) = 6.75\%$ for Pediatric ICU, and $(1-0.65)(1-0.74) = 9.1\%$ for Neonatal ICU. As shown in Table 2 (column 7), the actual simulation-generated mortality among admitted patients ($M^{\pi}_{\text{adm},j}$, Eq. 16) closely reproduces the registry targets across all four pools (e.g., 25.3%–26.1% at $\rho = 1.0$ and 25.5%–30.2% at $\rho = 1.6$ for Adult ICU). The boarding deterioration rate $\alpha^j_{\text{det}} \in [0.15, 0.20]$ day^{-1} is derived from the Brazilian cohort study of Cardoso et al. [12], which estimated an additional mortality hazard of ≈ 1.5% per hour of pre-ICU delay (≈ 0.36 day^{-1} on total mortality, corresponding to a decay rate of $\alpha_{\text{det}} \approx 0.18$ day^{-1} on the saveable survival margin $P^i_{\text{in},0} - P^i_{\text{out}}$, so that a 48-hour boarding delay erodes $1 - e^{-0.36} \approx 30\%$ of the initial ICU benefit).

(III) Assumed Counterfactual and Structural Parameters: The individual heterogeneity widths of P^i_{out} and g_i, the maximum waiting tolerance $W^i_{\max} \sim U(0.5, 2.5)$ days (representing the 12–60 hour clinical window documented in emergency boarding studies before stabilization, transfer, or death), the baseline severity–LOS coupling $\beta = 1.2$, and the baseline reneging payoff factor $\theta_{\text{ren}} = 1.0$ are modeling assumptions. Because P^i_{out} spans a width of 0.40 in each pool, setting $\beta = 1.2$ in Eq. (4) yields LOS multipliers ranging from $1 + 1.2(-0.20) = 0.76$ for the mildest arrivals to $1 + 1.2(+0.20) = 1.24$ for the sickest arrivals—a $1.24/0.76 = 1.63\times$ variation in expected stay across all pools. Because these structural parameters are not directly measured, we subject $\beta \in [-0.6, 2.4]$, $\alpha_{\text{det}} \in [0.05, 0.50]$ day^{-1}, $\theta_{\text{ren}} \in \{1.0, 0.5, 0.0\}$, and $\sigma_{\text{noise}} \in [0, 0.20]$ to comprehensive sensitivity analyses in Section 5.

4.2. Effective Traffic Intensity and Simulation Protocol

Because Eq. (4) centers the severity multiplier around $\mathbb{E}[1 - P^j_{\text{out}}]$ and the 0.35 floor is never active for $|\beta| \le 2.4$, the unconditional mean LOS in each pool j is exactly $\mathbb{E}[\text{LOS}_i] = \mu_{\text{los},j}$. By Little's Law, to achieve a uniform effective traffic intensity ρ across all four specialized units, the Poisson arrival rate for pool $j \in J$ is set to:

$$\lambda_j(\rho) = \frac{\rho \cdot C_j}{\mu_{\text{los},j}}, \qquad \forall j \in J \tag{22}$$

Thus, the nominal parameter ρ equals the true effective traffic intensity $\rho_{\text{eff}} = \lambda_j \mathbb{E}[\text{LOS}_j]/C_j = \rho$ in every pool. Over the $T = 60$-day evaluation window, the total expected arrival volume across all 55 beds is $60 \sum_{j \in J} \lambda_j(\rho) = 468.2\,\rho$ patients (yielding on average 329.9, 467.6, 605.2, and 744.7 evaluated arrivals at $\rho \in \{0.70, 1.00, 1.30, 1.60\}$, respectively).

Each replication simulates $T_{\text{warm}} = 15$ warm-up days followed by $T = 60$ evaluation days, continuing execution after day 75 until all patients arriving in $[15, 75)$ complete service or renege. To guarantee zero cross-contamination between random inputs under Common Random Numbers, SHA-256 hash-derived seeds (stable_seed) initialize five independent Mersenne Twister streams per replication: (1) inter-arrival times, (2) clinical survival attributes, (3) Log-Normal LOS realizations, (4) uniform survival outcomes u_i, and (5) clinical observation noise ε_i. Across all tables and figures, policies are reported in a fixed canonical order (FCFS, MSF, LSF, MGS, CMU-MGS, AUSTERE), and families of statistical comparisons are adjusted using the Holm–Bonferroni step-down procedure (p_{Holm}).

5. Results and Discussion

5.1. Overall Performance and LP Optimality Gaps

Table 3 reports the steady-state operational, admitted mortality, and survival metrics across all four congestion regimes ($\rho \in \{0.70, 1.00, 1.30, 1.60\}$), alongside both the Duration-Expectation LP Bound ($S^{\text{exp}}_{\text{UB}}$, using $\mathbb{E}[\text{LOS}_i]$) and the Hindsight LP Bound ($S^{\text{real}}_{\text{UB}}$, using realized LOS_i). Figure 1 illustrates the trajectory of expected survivors across traffic intensities relative to both LP upper bounds.

Several key operational patterns emerge from Table 3:

Table 3: Dynamic simulation performance across congestion regimes (60-day steady-state window after 15-day warm-up, 100 CRN replications; mean ± standard error) in fixed policy order, compared against the Duration-Expectation LP Bound ($\mathbb{E}[\text{LOS}_i]$) and Hindsight LP Bound (realized LOS_i).

Regime (ρ)	Policy	Total Arrivals	Admitted	Reneging Rate	Admitted Wait (h)	Reneged Wait (h)	Admitted LOS (d)	Bed Occupancy	Admitted Mort.	Realized Survivors	Expected Survivors
0.70	FCFS	329.9 ± 1.7	325.4 ± 1.6	1.3% ± 0.1%	0.74 ± 0.05	20.5 ± 0.8	7.04 ± 0.03	69.0% ± 0.4%	14.3% ± 0.0%	281.2 ± 1.5	281.44 ± 1.40
	MSF	329.9 ± 1.7	325.1 ± 1.6	1.4% ± 0.1%	0.61 ± 0.04	24.0 ± 0.9	7.04 ± 0.03	68.9% ± 0.4%	14.3% ± 0.0%	281.3 ± 1.5	281.59 ± 1.40
	LSF	329.9 ± 1.7	325.3 ± 1.6	1.4% ± 0.1%	0.56 ± 0.03	24.8 ± 0.9	7.04 ± 0.03	68.9% ± 0.4%	14.2% ± 0.0%	281.2 ± 1.5	281.49 ± 1.40
	MGS	329.9 ± 1.7	325.2 ± 1.6	1.4% ± 0.1%	0.59 ± 0.03	24.3 ± 0.9	7.04 ± 0.03	68.9% ± 0.4%	14.2% ± 0.0%	**281.3 ± 1.5**	281.65 ± 1.41
	CMU-MGS	329.9 ± 1.7	325.1 ± 1.6	1.4% ± 0.1%	0.56 ± 0.03	24.7 ± 0.9	7.04 ± 0.03	68.9% ± 0.4%	14.2% ± 0.0%	281.2 ± 1.5	281.60 ± 1.40
	AUSTERE	329.9 ± 1.7	325.4 ± 1.6	1.3% ± 0.1%	0.66 ± 0.04	25.7 ± 1.0	7.04 ± 0.03	68.9% ± 0.4%	14.3% ± 0.0%	281.2 ± 1.5	**281.67 ± 1.40**
	LP Bound ($\mathbb{E}[\text{LOS}]$)	329.9 ± 1.7	328.2 ± 1.6	—	—	—	—	—	—	—	*282.80 ± 1.43*
	Hindsight LP (Real. LOS)	329.9 ± 1.7	328.8 ± 1.7	—	—	—	—	—	—	—	*283.03 ± 1.45*
1.00	FCFS	467.6 ± 1.9	421.0 ± 1.5	9.9% ± 0.3%	5.65 ± 0.13	23.5 ± 0.2	7.06 ± 0.03	89.4% ± 0.2%	15.7% ± 0.1%	379.0 ± 1.5	378.51 ± 1.30
	MSF	467.6 ± 1.9	417.0 ± 1.5	10.7% ± 0.3%	4.17 ± 0.09	27.4 ± 0.2	7.10 ± 0.03	89.1% ± 0.2%	15.5% ± 0.1%	380.9 ± 1.6	380.70 ± 1.32
	LSF	467.6 ± 1.9	420.1 ± 1.5	10.1% ± 0.3%	3.57 ± 0.07	28.4 ± 0.2	7.03 ± 0.03	88.7% ± 0.2%	14.9% ± 0.0%	380.8 ± 1.6	379.94 ± 1.31
	MGS	467.6 ± 1.9	417.5 ± 1.5	10.6% ± 0.3%	3.87 ± 0.09	28.1 ± 0.2	7.09 ± 0.03	89.0% ± 0.2%	15.3% ± 0.0%	381.9 ± 1.6	381.33 ± 1.33
	CMU-MGS	467.6 ± 1.9	418.1 ± 1.4	10.5% ± 0.3%	3.68 ± 0.07	28.4 ± 0.2	7.07 ± 0.03	88.9% ± 0.2%	15.1% ± 0.0%	**382.2 ± 1.6**	**381.46 ± 1.32**
	AUSTERE	467.6 ± 1.9	419.1 ± 1.5	10.3% ± 0.3%	4.41 ± 0.10	30.1 ± 0.2	7.08 ± 0.03	89.2% ± 0.2%	15.3% ± 0.0%	381.7 ± 1.6	381.21 ± 1.33
	LP Bound ($\mathbb{E}[\text{LOS}]$)	467.6 ± 1.9	440.0 ± 1.3	—	—	—	—	—	—	—	*390.99 ± 1.37*
	Hindsight LP (Real. LOS)	467.6 ± 1.9	452.1 ± 1.6	—	—	—	—	—	—	—	*394.83 ± 1.48*
1.30	FCFS	605.2 ± 2.5	453.3 ± 1.4	25.0% ± 0.3%	12.68 ± 0.14	24.7 ± 0.1	7.08 ± 0.02	96.8% ± 0.1%	17.5% ± 0.1%	447.8 ± 1.7	449.11 ± 1.47
	MSF	605.2 ± 2.5	442.5 ± 1.4	26.8% ± 0.3%	8.42 ± 0.10	29.1 ± 0.1	7.27 ± 0.03	96.6% ± 0.1%	17.0% ± 0.0%	455.1 ± 1.7	456.59 ± 1.54
	LSF	605.2 ± 2.5	456.0 ± 1.3	24.6% ± 0.3%	6.91 ± 0.08	29.7 ± 0.1	6.99 ± 0.02	96.1% ± 0.1%	15.3% ± 0.0%	454.1 ± 1.7	455.50 ± 1.52
	MGS	605.2 ± 2.5	444.8 ± 1.4	26.4% ± 0.3%	7.56 ± 0.08	29.6 ± 0.1	7.22 ± 0.03	96.5% ± 0.1%	16.3% ± 0.0%	457.4 ± 1.8	458.94 ± 1.57
	CMU-MGS	605.2 ± 2.5	448.2 ± 1.4	25.9% ± 0.3%	7.13 ± 0.08	29.9 ± 0.1	7.16 ± 0.03	96.4% ± 0.1%	15.9% ± 0.0%	**457.7 ± 1.8**	**459.59 ± 1.55**
	AUSTERE	605.2 ± 2.5	449.0 ± 1.4	25.7% ± 0.3%	8.79 ± 0.10	31.7 ± 0.1	7.16 ± 0.03	96.7% ± 0.1%	16.3% ± 0.0%	457.6 ± 1.7	458.69 ± 1.55
	LP Bound ($\mathbb{E}[\text{LOS}]$)	605.2 ± 2.5	485.0 ± 0.8	—	—	—	—	—	—	—	*480.32 ± 1.57*
	Hindsight LP (Real. LOS)	605.2 ± 2.5	543.0 ± 1.8	—	—	—	—	—	—	—	*495.53 ± 1.86*
1.60	FCFS	744.7 ± 2.5	462.1 ± 1.4	37.9% ± 0.3%	18.27 ± 0.13	25.5 ± 0.1	7.08 ± 0.02	98.9% ± 0.1%	18.8% ± 0.0%	512.6 ± 1.6	513.42 ± 1.37
	MSF	744.7 ± 2.5	444.5 ± 1.4	40.2% ± 0.3%	10.99 ± 0.07	29.8 ± 0.1	7.39 ± 0.03	98.8% ± 0.1%	18.2% ± 0.0%	527.4 ± 1.7	527.54 ± 1.45
	LSF	744.7 ± 2.5	472.0 ± 1.4	36.6% ± 0.3%	8.80 ± 0.06	30.5 ± 0.1	6.94 ± 0.02	98.5% ± 0.1%	15.1% ± 0.0%	525.7 ± 1.7	525.74 ± 1.42
	MGS	744.7 ± 2.5	446.8 ± 1.4	39.9% ± 0.3%	9.68 ± 0.06	30.4 ± 0.1	7.34 ± 0.02	98.8% ± 0.1%	16.9% ± 0.0%	530.2 ± 1.8	530.61 ± 1.46
	CMU-MGS	744.7 ± 2.5	454.2 ± 1.4	38.9% ± 0.3%	8.99 ± 0.05	30.7 ± 0.1	7.23 ± 0.02	98.7% ± 0.1%	16.1% ± 0.0%	**531.6 ± 1.7**	**532.09 ± 1.46**
	AUSTERE	744.7 ± 2.5	454.0 ± 1.4	39.0% ± 0.3%	11.38 ± 0.08	32.7 ± 0.1	7.24 ± 0.02	98.9% ± 0.1%	16.7% ± 0.0%	530.5 ± 1.7	530.27 ± 1.43
	LP Bound ($\mathbb{E}[\text{LOS}]$)	744.7 ± 2.5	495.7 ± 0.6	—	—	—	—	—	—	—	*559.17 ± 1.44*
	Hindsight LP (Real. LOS)	744.7 ± 2.5	614.4 ± 1.2	—	—	—	—	—	—	—	*590.47 ± 1.67*

(1) Consistency with Little's Law and Empirical Mortality Targets: At $\rho = 0.70$, only 1.3%–1.4% of arrivals renege, yielding a steady-state occupancy of 68.9%–69.0% (matching the theoretical $M/G/c + G$ load $\rho(1 - r_{\text{ren}}) \approx 0.70 \times 0.986 = 69.0\%$). At $\rho = 1.00$, stochastic fluctuations under finite patience induce a 9.9%–10.7% reneging rate and 88.7%–89.4% bed utilization, while at $\rho = 1.60$ the hospital operates near full saturation (98.5%–98.9% occupancy). Across all regimes, the overall hospital mortality among admitted patients (14.2%–18.8%, weighted across the four specialty pools) and the disaggregated pool mortalities (Table 2) match the AMIB/Epimed national registry benchmarks.

(2) Bed Turnover vs. Clinical Severity across Policies: The "Admitted LOS" column reflects the severity–duration coupling of Eq. (4). At $\rho = 1.60$, MSF prioritizes the sickest candidates ($1 - P^i_{\text{out}}$), resulting in the longest mean admitted stay (7.39 days) and the lowest admission volume (444.5 patients, 40.2% reneging). Conversely, LSF favors milder patients with high P^i_{in}, achieving the shortest mean stay (6.94 days) and admitting the highest number of patients (472.0, 36.6% reneging), but saving fewer lives than MGS and CMU-MGS because many of its admitted patients already had high ward survival P^i_{out}. By normalizing marginal gain by expected stay, CMU-MGS admits 7.4 more patients than greedy MGS at $\rho = 1.60$ (454.2 vs. 446.8) with a shorter mean stay (7.23 vs. 7.34 days), directly translating faster bed turnover into higher overall survival (532.09 vs. 530.61).

(3) Tightening the LP Upper Bound (Value of Clairvoyance vs. Online Optimality Gap): As shown in Table 3 and Figure 1, the Hindsight LP Bound ($S^{\text{real}}_{\text{UB}}$, which knows realized individual stays LOS_i) achieves 283.03, 394.83, 495.53, and 590.47 expected survivors across $\rho \in \{0.7, 1.0, 1.3, 1.6\}$, corresponding to a pessimistic gap $(S^{\text{real}}_{\text{UB}} - S^{\text{CMU-MGS}}_{\text{exp}})/S^{\text{real}}_{\text{UB}}$ of 0.51%, 3.39%, 7.25%, and 9.89%. However, at $\rho = 1.60$, $S^{\text{real}}_{\text{UB}}$ admits 614.4 patients (compared to 454.2 under CMU-MGS) because hindsight knowledge of log-normal realizations allows the LP to selectively cherry-pick patients who happen to draw ultra-short realized stays ($\text{LOS}_i \ll \mathbb{E}[\text{LOS}_i]$). When we remove duration clairvoyance via the Duration-Expectation LP Bound ($S^{\text{exp}}_{\text{UB}}$, which uses $\mathbb{E}[\text{LOS}_i]$), LP admissions drop to 495.7 and the upper bound tightens to 282.80, 390.99, 480.32, and 559.17 survivors across $\rho \in \{0.7, 1.0, 1.3, 1.6\}$. Against $S^{\text{exp}}_{\text{UB}}$, the optimality gap of CMU-MGS drops by more than half under severe surge: 0.42% at $\rho = 0.70$, 2.44% at $\rho = 1.00$, 4.32% at $\rho = 1.30$, and **4.84**% at $\rho = 1.60$ (with the remaining 4.84% gap attributable to future arrival-time knowledge, fractional LP splitting, and empty initial bed relaxation at T_{warm}).

5.2. Paired Statistical Comparisons Over the FCFS Baseline

Table 4 and Figure 2 report the Common Random Numbers paired survival differences $\Delta S^{\pi}_r = S^{\pi}_{\text{exp},r} - S^{\text{FCFS}}_{\text{exp},r}$ across all 20 policy-regime pairs in fixed policy order, adjusted for multiple testing via the Holm–Bonferroni procedure.

At low traffic intensity ($\rho = 0.70$), because variance reduction under CRN is substantial, small differences in how occasional bursts of arrivals are ordered produce statistically detectable differences for MSF, MGS, CMU-MGS, and AUSTERE ($p_{\text{Holm}} < 0.001$), whereas LSF is statistically indistinguishable from FCFS ($p_{\text{Holm}} = 0.200$). In

Table 4: Paired expected survival differences relative to the FCFS hospital baseline across 100 CRN replications in fixed policy order (mean ± SE, 95% paired CI, relative gain, paired t-statistic, and Holm–Bonferroni adjusted p-value across 20 tests).

Regime (ρ)	Policy vs. FCFS	Paired Difference (ΔS)	95% Paired CI	Relative Effect	t-statistic	p_{Holm}
0.70	MSF vs. FCFS	+0.15 ± 0.03	[+0.08, +0.21]	+0.05%	4.30	< 0.001
	LSF vs. FCFS	+0.05 ± 0.04	[−0.03, +0.13]	+0.02%	1.29	0.200
	MGS vs. FCFS	+0.21 ± 0.04	[+0.13, +0.29]	+0.07%	4.86	< 0.001
	CMU-MGS vs. FCFS	+0.16 ± 0.04	[+0.08, +0.24]	+0.06%	4.09	< 0.001
	AUSTERE vs. FCFS	**+0.23 ± 0.04**	[+0.15, +0.31]	+0.08%	5.66	< 0.001
1.00	MSF vs. FCFS	+2.18 ± 0.15	[+1.89, +2.47]	+0.58%	14.66	< 0.001
	LSF vs. FCFS	+1.43 ± 0.15	[+1.14, +1.72]	+0.38%	9.57	< 0.001
	MGS vs. FCFS	+2.82 ± 0.16	[+2.50, +3.14]	+0.75%	17.35	< 0.001
	CMU-MGS vs. FCFS	**+2.94 ± 0.17**	[+2.61, +3.28]	+0.78%	17.36	< 0.001
	AUSTERE vs. FCFS	+2.70 ± 0.17	[+2.37, +3.02]	+0.71%	16.32	< 0.001
1.30	MSF vs. FCFS	+7.48 ± 0.30	[+6.90, +8.07]	+1.67%	25.19	< 0.001
	LSF vs. FCFS	+6.39 ± 0.30	[+5.81, +6.97]	+1.42%	21.51	< 0.001
	MGS vs. FCFS	+9.83 ± 0.30	[+9.23, +10.43]	+2.19%	32.28	< 0.001
	CMU-MGS vs. FCFS	**+10.48 ± 0.29**	[+9.92, +11.05]	+2.33%	36.31	< 0.001
	AUSTERE vs. FCFS	+9.58 ± 0.31	[+8.98, +10.18]	+2.13%	31.24	< 0.001
1.60	MSF vs. FCFS	+14.12 ± 0.38	[+13.38, +14.86]	+2.75%	37.30	< 0.001
	LSF vs. FCFS	+12.32 ± 0.42	[+11.51, +13.14]	+2.40%	29.60	< 0.001
	MGS vs. FCFS	+17.19 ± 0.41	[+16.39, +17.99]	+3.35%	42.09	< 0.001
	CMU-MGS vs. FCFS	**+18.66 ± 0.42**	[+17.85, +19.48]	+3.64%	44.79	< 0.001
	AUSTERE vs. FCFS	+16.85 ± 0.42	[+16.02, +17.67]	+3.28%	40.01	< 0.001

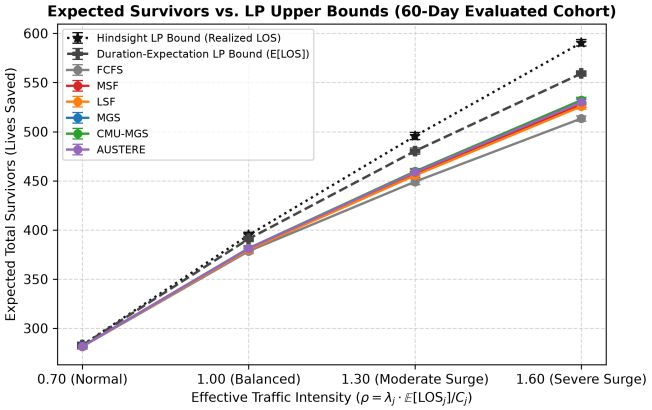


Figure 1: Expected total survivors across online triage policies compared against the Hindsight LP Bound (realized LOS_i) and the tighter Duration-Expectation LP Bound ($\mathbb{E}[\text{LOS}_i]$) across effective traffic intensities $\rho \in \{0.7, 1.0, 1.3, 1.6\}$ (95% confidence intervals).

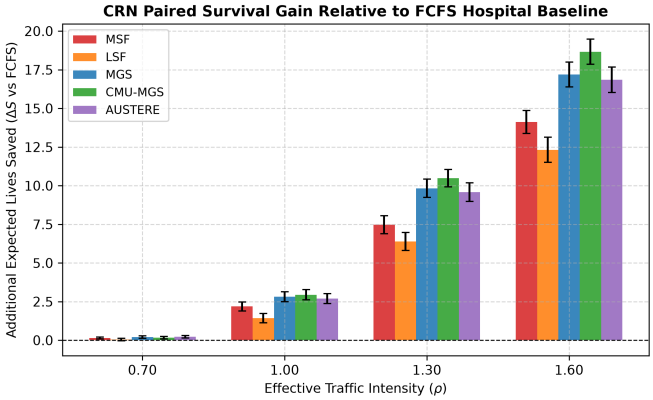


Figure 2: Paired incremental expected lives saved (ΔS) over the FCFS baseline across traffic intensities $\rho \in \{0.7, 1.0, 1.3, 1.6\}$ (95% paired confidence intervals).

practical operational terms, however, the effect size at $\rho = 0.70$ is negligible across all rules ($\Delta S \leq 0.23$ lives over 60 days, or $< 0.08\%$ relative survival gain).

As traffic intensity reaches moderate ($\rho = 1.30$) and severe ($\rho = 1.60$) scarcity, clinically meaningful differences emerge. At $\rho = 1.60$, CMU-MGS saves $+18.66 \pm 0.42$ more expected lives than FCFS (+3.64%, $p_{\text{Holm}} < 0.001$), outperforming greedy MGS ($+17.19 \pm 0.41$), AUSTERE ($+16.85 \pm 0.42$), MSF ($+14.12 \pm 0.38$), and LSF ($+12.32 \pm 0.42$). In a direct paired test at $\rho = 1.60$ under the baseline parameter $\beta = 1.2$, CMU-MGS outperforms greedy MGS by $+1.48 \pm 0.26$ expected lives ($t = 5.71, p < 0.001$), confirming that normalizing marginal benefit by expected stay improves survival even under moderate severity–LOS coupling.

5.3. Disaggregated Waiting Times, Per-Pool Performance, and Overflow Ablation

Figure 3 displays the mean waiting time of admitted patients (Panel A) and the queue reneging rate (Panel B) across congestion levels. Under chronological FCFS at $\rho = 1.60$, every newly arriving candidate must wait behind the entire backlog of earlier arrivals, driving the mean waiting time of admitted patients to 18.27 hours and exposing all admitted patients to substantial exponential survival decay under Eq. (2). Under non-FCFS priority rules, high-priority candidates bypass the queue upon bed

release, reducing the mean wait of admitted patients to 8.80–11.38 hours while lower-priority patients remain at the back of the queue until reaching their reneging deadline (29.8–32.7 hours). Notice that because the heuristic AUSTERE rule (Eq. 12) boosts the score of candidates nearing their deadline, it admits slightly longer-waiting candidates (11.38 hours vs. 8.99 hours under CMU-MGS) and achieves slightly lower total survival than pure CMU-MGS (530.27 vs. 532.09 at $\rho = 1.60$).

Table 5 reports the disaggregated admissions and occupancy rates across the four specialized pools across all four traffic intensities $\rho \in \{0.70, 1.00, 1.30, 1.60\}$, together with the paired ablation of the hierarchical adult–coronary overflow mechanism ($\Delta S_{\text{overflow}} = S_{\text{exp}}^{\text{overflow ON}} - S_{\text{exp}}^{\text{overflow OFF}}$), with p-values adjusted across all 12 ablation tests via the Holm–Bonferroni procedure.

The ablation study in Table 5 reveals a classic queueing pooling phenomenon. Cross-unit overflow between the 15-bed Adult ICU and 10-bed Coronary Unit is most impactful at balanced traffic ($\rho = 1.00$), where asynchronous random bursts temporarily saturate one unit while the partner unit has idle beds: enabling overflow admits +11.89 additional patients and saves $+4.27 \pm 0.16$ expected lives under CMU-MGS ($t = 26.97, p_{\text{Holm}} < 0.001$) and $+4.55 \pm 0.20$ lives under FCFS ($p_{\text{Holm}} < 0.001$). Statistically significant gains also occur at $\rho = 0.70$ (+1.65 to +1.77 lives, $p_{\text{Holm}} < 0.001$) and $\rho = 1.30$ (+1.42 to +2.41 lives, $p_{\text{Holm}} < 0.001$). Conversely, at extreme overload ($\rho = 1.60$), both the adult and coronary pools are continuously saturated (99.6%–99.8% occupancy) with persistent native queues, leaving almost no idle capacity for cross-unit sharing ($\Delta S = -0.09$ to +0.51, $p_{\text{Holm}} \geq 0.050$).

5.4. Sensitivity Analysis: Imperfect Information, Severity–LOS Regimes, and Reneging Payoffs

To evaluate how informational errors, negative or positive severity–LOS coupling, and pre-ICU mortality among reneging patients affect policy rankings, we conduct three sensitivity experiments at $\rho = 1.60$.

5.4.1. Robustness to Multi-Parameter Clinical Noise

As shown in Figure 4A, when prognostic probabilities and expected stay estimates are perturbed by simultaneous Gaussian noise, all five clinical policies degrade gracefully while remaining strictly and statistically superior to chronological FCFS (513.42 ± 1.37 survivors) across the entire noise range. At $\sigma_{\text{noise}} = 0.10$, the paired gains over FCFS remain highly significant for all five rules: MGS ($+14.89 \pm 0.35$, $t = 42.92$), CMU-MGS ($+13.88 \pm 0.39$, $t = 35.65$), MSF ($+12.83 \pm 0.37$, $t = 34.62$), AUSTERE ($+12.73 \pm 0.41$, $t = 31.24$), and LSF ($+11.85 \pm 0.41$, $t = 28.93$), all with $p_{\text{Holm}} < 0.001$. Even under severe simultaneous noise ($\sigma_{\text{noise}} = 0.20$), all five clinical policies maintain a large, statistically significant survival advantage over FCFS: MGS saves $+12.82 \pm 0.37$ lives (526.24 total, $t = 34.43$), CMU-MGS saves $+11.82 \pm 0.41$ lives (525.24 total, $t = 28.82$), MSF saves $+11.67 \pm 0.38$ lives (525.09 total, $t = 30.99$), AUSTERE saves $+11.15 \pm 0.34$ lives (524.57 total, $t = 32.61$), and LSF saves $+10.41 \pm 0.40$ lives (523.83 total, $t = 26.14$), all with $p_{\text{Holm}} < 0.001$. Notice that when $\sigma_{\text{noise}} \geq 0.10$, because CMU-MGS compounds estimation error in both its numerator ($\widetilde{\Delta P}_i$) and denominator ($\widetilde{\mathbb{E}}[\text{LOS}_i]$), unnormalized MGS edges ahead slightly (526.24 vs. 525.24 at $\sigma_{\text{noise}} = 0.20$), consistent with the theoretical findings of Argon and Ziya [22].

5.4.2. Direction and Strength of Severity–LOS Coupling ($\beta \in [-0.6, 2.4]$)

In intensive care medicine, the empirical relationship between physiological severity and length of stay is nonmonotonic: while higher severity among survivors requires longer mechanical ventilation ($\beta > 0$), patients with extreme severity who die early in the ICU often have truncated stays ($\beta < 0$) [19?]. Figure 4B maps the paired survival gain of CMU-MGS over MGS across 24 parameter cells spanning $\beta \in \{-0.6, 0.0, 0.6, 1.2, 1.8, 2.4\}$ and $\alpha_{\text{det}} \in \{0.05, 0.18, 0.35, 0.50\}$ day^{-1}, with significance asterisks indicating Holm–Bonferroni adjusted $p_{\text{Holm}} < 0.05$ across all 24 cells:

- Negative Coupling ($\beta = -0.6$, Early In-ICU Mortality of Severe Cases): When sicker patients have shorter expected stays, high marginal survival gain (ΔP_i) and short stay ($\mathbb{E}[\text{LOS}_i]$) are already aligned. Consequently, MGS and CMU-MGS select nearly identical patients, and their difference is statistically indistinguishable from zero across all deterioration rates ($\Delta S_{\text{CMU–MGS}} \in [-0.21, +0.35]$, $p_{\text{Holm}} = 1.00$), while both rules outperform FCFS by +18.9 to +37.2 lives.

- Zero or Weak Positive Coupling ($\beta \in \{0.0, 0.6\}$): At $\beta = 0.0$, $\mathbb{E}[\text{LOS}_i] = \mu_{\text{los},j}$ is constant within each pool, so CMU-MGS and MGS produce identical rankings ($\Delta S_{\text{CMU–MGS}} = 0.00$). At $\beta = 0.6$, differences remain small and exploratory ($\Delta S_{\text{CMU–MGS}} \in [-0.06, +0.67]$, $p_{\text{Holm}} = 1.00$).

- Moderate to Strong Positive Coupling ($\beta \in \{1.2, 1.8, 2.4\}$): When severe patients require longer stays, greedy MGS locks beds with long-stay admissions. At the baseline cell ($\beta = 1.2, \alpha_{\text{det}} = 0.18$), CMU-MGS saves $+1.42 \pm 0.34$ more lives than MGS ($t = 4.19, p_{\text{Holm}} < 0.001$). For $\beta = 1.8$ and $\beta = 2.4$, CMU-MGS significantly outperforms MGS across every deterioration rate ($p_{\text{Holm}} < 0.001$ in all 8 cells), saving +2.09 to +2.49 additional lives at $\beta = 1.8$ and +4.22 to +4.99 additional lives at $\beta = 2.4$.

5.4.3. Sensitivity to Reneging Survival Payoff ($\theta_{\text{ren}} \in \{1.0, 0.5, 0.0\}$)

In our baseline specification ($\theta_{\text{ren}} = 1.0$), every patient who reneges after reaching W_{max}^i receives the counterfactual ward survival probability P_{out}^i. However, because

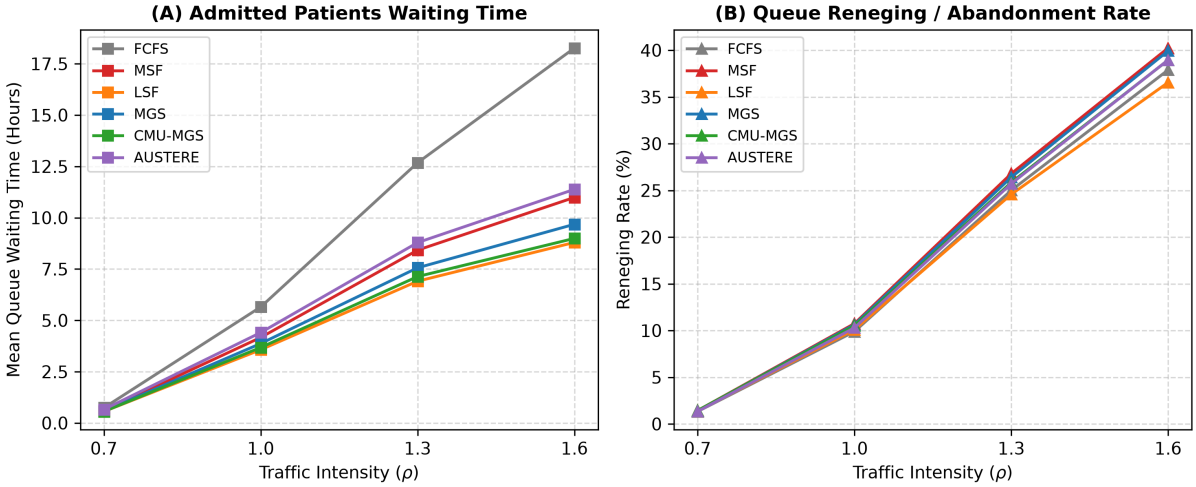


Figure 3: Queueing dynamics across traffic intensities ρ: (A) mean waiting time of admitted patients (hours); (B) percentage of arriving patients who renege before ICU admission.

Table 5: Disaggregated per-pool admissions and occupancy (mean over 100 replications) and paired ablation of adult–coronary overflow pooling (p_{Holm} adjusted across all 12 tests).

		Disaggregated Admissions / Arrivals (Pool Occupancy %)				Adult–Coronary Overflow Ablation (ON minus OFF)		
Regime (ρ)	Policy	Adult (15 beds)	Coronary (10 beds)	Pediatric (10 beds)	Neonatal (20 beds)	Δ Admissions	Δ Expected Survivors	Paired t (p_{Holm})
0.70	FCFS	97.1/97.5 (70.7%)	87.3/87.8 (68.7%)	67.6/70.1 (67.4%)	73.5/74.5 (68.6%)	+3.52 ± 0.27	+1.77 ± 0.11	t = 15.49 ($p_{\text{Holm}} < 0.001$)
	MGS	97.1/97.5 (70.7%)	87.2/87.8 (68.7%)	67.5/70.1 (67.2%)	73.4/74.5 (68.4%)	+3.96 ± 0.29	+1.66 ± 0.11	t = 15.70 ($p_{\text{Holm}} < 0.001$)
	CMU-MGS	97.1/97.5 (70.7%)	87.2/87.8 (68.7%)	67.4/70.1 (67.2%)	73.4/74.5 (68.4%)	+3.89 ± 0.29	+1.65 ± 0.11	t = 15.55 ($p_{\text{Holm}} < 0.001$)
1.00	FCFS	129.9/139.8 (92.5%)	112.8/124.3 (91.3%)	85.9/99.4 (86.4%)	92.4/104.1 (87.6%)	+11.21 ± 0.47	+4.55 ± 0.20	t = 22.68 ($p_{\text{Holm}} < 0.001$)
	MGS	128.8/139.8 (92.0%)	112.1/124.3 (91.0%)	84.8/99.4 (85.9%)	91.8/104.1 (87.3%)	+11.89 ± 0.43	+4.14 ± 0.16	t = 25.50 ($p_{\text{Holm}} < 0.001$)
	CMU-MGS	129.2/139.8 (91.9%)	112.0/124.3 (90.8%)	84.9/99.4 (85.7%)	91.9/104.1 (87.3%)	+11.89 ± 0.42	+4.27 ± 0.16	t = 26.97 ($p_{\text{Holm}} < 0.001$)
1.30	FCFS	139.9/177.5 (98.8%)	118.9/162.9 (98.5%)	94.6/130.6 (94.5%)	99.9/134.2 (95.5%)	+6.09 ± 0.51	+1.42 ± 0.19	t = 7.55 ($p_{\text{Holm}} < 0.001$)
	MGS	137.2/177.5 (98.5%)	116.9/162.9 (98.3%)	92.0/130.6 (94.5%)	98.8/134.2 (95.2%)	+7.56 ± 0.52	+1.98 ± 0.17	t = 11.40 ($p_{\text{Holm}} < 0.001$)
	CMU-MGS	138.8/177.5 (98.4%)	118.3/162.9 (98.0%)	92.2/130.6 (94.4%)	98.9/134.2 (95.2%)	+8.63 ± 0.59	+2.41 ± 0.20	t = 12.21 ($p_{\text{Holm}} < 0.001$)
1.60	FCFS	143.1/222.1 (99.8%)	118.5/198.1 (99.8%)	97.5/158.8 (97.9%)	103.0/165.7 (98.3%)	+1.13 ± 0.59	−0.09 ± 0.23	t = −0.40 ($p_{\text{Holm}} = 1.000$)
	MGS	139.9/222.1 (99.7%)	113.1/198.1 (99.7%)	94.1/158.8 (97.8%)	99.7/165.7 (98.1%)	+1.28 ± 0.61	+0.03 ± 0.22	t = 0.13 ($p_{\text{Holm}} = 1.000$)
	CMU-MGS	143.4/222.1 (99.7%)	115.5/198.1 (99.6%)	95.0/158.8 (97.8%)	100.3/165.7 (98.1%)	+2.26 ± 0.63	+0.51 ± 0.22	t = 2.39 ($p_{\text{Holm}} = 0.050$)

$W^i_{\max}$ encompasses pre-ICU death as well as ward stabilization or transfer, attributing the full P^i_{out} to all abandonments could favor policies with higher reneging rates (such as MSF and MGS) over high-throughput policies (such as LSF and CMU-MGS). Table 6 evaluates all six policies at $\rho = 1.60$ across 100 replications under three reneging survival retention factors: $\theta_{\text{ren}} = 1.0$ (baseline ward survival), $\theta_{\text{ren}} = 0.5$ (50% survival penalty upon reneging, representing a 50/50 mixture of ward transfer and pre-ICU death), and $\theta_{\text{ren}} = 0.0$ (100% pre-ICU mortality upon reneging).

Table 6 confirms the referee's structural insight and reveals three important operational results:

(1) Penalization of High-Reneging Policies (MSF and Unnormalized MGS): When reneging patients receive only $0.5 \times P^i_{\text{out}}$ ($\theta_{\text{ren}} = 0.5$), MSF—which has the highest reneging rate (40.2%) because it locks beds with long-stay severe patients—sees its advantage over FCFS collapse from +14.12 lives to +1.45 lives (and becomes strictly worse than FCFS, −11.22 lives, when $\theta_{\text{ren}} = 0.0$).

(2) Surge of LSF and Expansion of the CMU-MGS Advantage over MGS: Because LSF admits shorter-stay patients with high P^i_{in}, its lower reneging rate (36.6%) causes it to overtake greedy MGS at $\theta_{\text{ren}} = 0.5$ (463.35 vs. 458.01 survivors). Simultaneously, because CMU-MGS incorporates the net benefit $P^i_{\text{in}}(w) - \theta_{\text{ren}} P^i_{\text{out}}$ divided by $\mathbb{E}[\text{LOS}_i]$ (Eq. 11), it naturally increases bed turnover as θ_{ren} decreases, expanding its admissions from 454.2 ($\theta_{\text{ren}} = 1.0$) to 475.5 ($\theta_{\text{ren}} = 0.5$) and 478.9 ($\theta_{\text{ren}} = 0.0$). Consequently, the paired advantage of CMU-MGS over greedy MGS increases more than threefold at $\theta_{\text{ren}} = 0.5$, rising from +1.48 ± 0.26 lives to **+5.41 ± 0.50** lives ($t = 10.77$, $p_{\text{Holm}} < 0.001$).

(3) Invariance of the Top Policy: At $\theta_{\text{ren}} = 0.0$, where every abandonment is a death, $P^i_{\text{in}}(w) - \theta_{\text{ren}} P^i_{\text{out}} = P^i_{\text{in}}(w)$, so MGS and LSF become mathematically identical (400.96 survivors), whereas CMU-MGS ($P^i_{\text{in}}(w)/\mathbb{E}[\text{LOS}_i]$) remains the top-ranked policy (404.87 survivors), outperforming both MGS/LSF by +3.91 ± 0.67 lives ($p_{\text{Holm}} < 0.001$) and FCFS by +29.84 ± 0.86 lives.

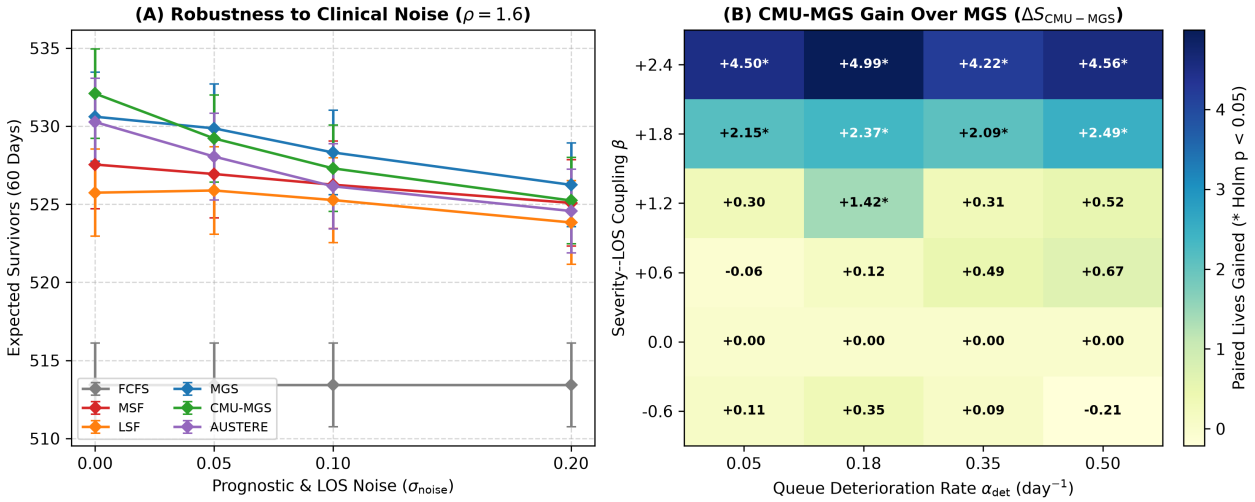


Figure 4: Sensitivity analysis at severe congestion ($\rho = 1.60$): (A) expected survivors across 100 replications under simultaneous observation noise $\sigma_{\text{noise}} \in \{0.00, 0.05, 0.10, 0.20\}$ affecting P_{out}, $P_{\text{in},0}$, and $\mathbb{E}[\text{LOS}]$; (B) 24-cell parametric heatmap of the paired survival gain of CMU-MGS over greedy MGS ($\Delta S_{\text{CMU–MGS}}$) across severity–LOS coupling $\beta \in \{-0.6, 0.0, 0.6, 1.2, 1.8, 2.4\}$ and queue deterioration rate $\alpha_{\text{det}} \in \{0.05, 0.18, 0.35, 0.50\}$ day^{-1} ($*$ indicates $p_{\text{Holm}} < 0.05$ across the 24 tests).

Table 6: Sensitivity of policy performance and rankings at $\rho = 1.60$ (100 CRN replications) to the reneging survival retention factor $\theta_{\text{ren}} \in \{1.0, 0.5, 0.0\}$, where reneging patients survive with probability $\theta_{\text{ren}} P^i_{\text{out}}$.

Reneging Payoff (θ_{ren})	Policy	Admitted Patients	Reneging Rate	Expected Survivors	Paired Gain vs. FCFS (ΔS)	CMU-MGS Advantage
$\theta_{\text{ren}} = 1.0$ (Full P^i_{out})	FCFS	462.1 ± 1.4	37.9%	513.42 ± 1.37	Baseline	+18.66 ± 0.42 ($p_{\text{Holm}} < 0.001$)
	MSF	444.5 ± 1.4	40.2%	527.54 ± 1.45	+14.12 ± 0.38	+4.54 ± 0.27 ($p_{\text{Holm}} < 0.001$)
	LSF	472.0 ± 1.4	36.6%	525.74 ± 1.42	+12.32 ± 0.42	+6.34 ± 0.33 ($p_{\text{Holm}} < 0.001$)
	MGS	446.8 ± 1.4	39.9%	530.61 ± 1.46	+17.19 ± 0.41	+1.48 ± 0.26 ($p_{\text{Holm}} < 0.001$)
	CMU-MGS	454.2 ± 1.4	38.9%	**532.09 ± 1.46**	**+18.66 ± 0.42**	Rank 1
	AUSTERE	454.0 ± 1.4	39.0%	530.27 ± 1.43	+16.85 ± 0.42	+1.82 ± 0.19 ($p_{\text{Holm}} < 0.001$)
$\theta_{\text{ren}} = 0.5$ ($0.5 \times P^i_{\text{out}}$)	FCFS	462.1 ± 1.4	37.9%	444.23 ± 1.13	Baseline	+19.19 ± 0.63 ($p_{\text{Holm}} < 0.001$)
	MSF	444.5 ± 1.4	40.2%	445.67 ± 1.13	+1.45 ± 0.60	+17.74 ± 0.59 ($p_{\text{Holm}} < 0.001$)
	LSF	472.0 ± 1.4	36.6%	463.35 ± 1.17	+19.12 ± 0.62	+0.07 ± 0.43 ($p_{\text{Holm}} = 0.878$)
	MGS	456.5 ± 1.4	38.6%	458.01 ± 1.14	+13.78 ± 0.59	+5.41 ± 0.50 ($p_{\text{Holm}} < 0.001$)
	CMU-MGS	475.5 ± 1.4	36.1%	**463.42 ± 1.13**	**+19.19 ± 0.63**	Rank 1
	AUSTERE	477.7 ± 1.4	35.8%	460.71 ± 1.13	+16.48 ± 0.65	+2.71 ± 0.42 ($p_{\text{Holm}} < 0.001$)
$\theta_{\text{ren}} = 0.0$ (Zero Payoff)	FCFS	462.1 ± 1.4	37.9%	375.03 ± 1.18	Baseline	+29.84 ± 0.86 ($p_{\text{Holm}} < 0.001$)
	MSF	444.5 ± 1.4	40.2%	363.81 ± 1.16	−11.22 ± 0.89	+41.06 ± 0.88 ($p_{\text{Holm}} < 0.001$)
	LSF	472.0 ± 1.4	36.6%	400.96 ± 1.24	+25.93 ± 0.86	+3.91 ± 0.67 ($p_{\text{Holm}} < 0.001$)
	MGS	472.0 ± 1.4	36.6%	400.96 ± 1.24	+25.93 ± 0.86	+3.91 ± 0.67 ($p_{\text{Holm}} < 0.001$)
	CMU-MGS	478.9 ± 1.4	35.6%	**404.87 ± 1.20**	**+29.84 ± 0.86**	Rank 1
	AUSTERE	480.7 ± 1.4	35.4%	403.90 ± 1.21	+28.87 ± 0.88	+0.97 ± 0.56 ($p_{\text{Holm}} = 0.086$)

6. Limitations and Ethical Considerations

Translating operations research models into clinical triage guidelines requires explicit recognition of modeling boundaries and bioethical constraints:

(1) Registry-Level Calibration vs. Patient-Level Electronic Health Records: While our bed capacities are taken directly from DATASUS/CNES for HU-UFMA and specialty-level LOS and admitted mortality rates are calibrated to match national registries (AMIB/Epimed and RBPN), individual counterfactual baseline survival (P^i_{out}) and ICU efficacy (g_i) are unobserved in admitted-only registries and are modeled parametrically (Table 2). Future clinical deployment requires fitting joint survival–duration copulas to de-identified patient-level electronic health records (EHR) that include both admitted and refused/ward-managed referrals.

(2) Prognostic Uncertainty and Algorithmic Equity: Efficiency-oriented policies (MGS, CMU-MGS, AUSTERE) rely on prognostic survival and LOS estimates. In clinical practice, severity scores such as SOFA and SAPS 3 exhibit calibration drift across demographic and socioeconomic groups [5, 2]. Furthermore, pure $c\mu$ prioritization systematically penalizes patients with chronic comorbidities or severe multi-organ dysfunction whose expected recovery time $\mathbb{E}[\text{LOS}_i]$ is protracted. Triage committees should therefore embed index scores within multi-principle frameworks [5] that incorporate equity safeguards, age/life-cycle considerations, and periodic clinical reassessment rather than delegating admission decisions to an automated score

alone.

(3) Operational Scope: Our discrete-event engine assumes stationary Poisson arrival rates within each 60-day regime, instantaneous bed turnaround upon discharge (omitting 2–4 hour sanitization delays), and no premature step-down discharge (demand-driven early discharge as in Chan et al. [9]). Incorporating non-homogeneous epidemic wave arrivals and dynamic step-down transfers represents a natural extension.

7. Conclusions

This paper developed a continuous-time discrete-event simulation and retrospective LP benchmarking framework for dynamic multi-type ICU bed allocation under resource scarcity. Grounded in the 55-bed critical care structure of HU-UFMA (DATASUS/CNES) and calibrated to match specialty-specific LOS and hospital mortality targets from Brazilian intensive care registries, our study yields four main conclusions:

(1) Value of Dynamic Triage and Tightened LP Optimality Gap: While admission rules have negligible practical impact when capacity is ample ($\rho = 0.70$, relative effect $< 0.08\%$), priority indices that account for marginal survival benefit and bed-time consumption (CMU-MGS) save $+18.66 \pm 0.42$ more expected lives than FCFS over a 60-day surge ($\rho = 1.60$, $+3.64\%$, $p_{\text{Holm}} < 0.001$) and cut admitted patient waiting times by half (8.99 vs. 18.27 hours). While the pessimistic gap to a clairvoyant hindsight LP bound (which cherry-picks ultra-short realized stays) is 9.89%, evaluating CMU-MGS against a duration-expectation LP bound ($\mathbb{E}[\text{LOS}_i]$) shows that the online policy comes within 4.84% of the offline optimum.

(2) Regime-Dependent Advantage of $c\mu$ Normalization: Our 24-cell sensitivity analysis across $\beta \in [-0.6, 2.4]$ demonstrates that greedy marginal survival triage (MGS) suffices when length of stay is negatively or weakly correlated with severity ($\beta \leq 0.6$). However, when severe patients require longer ICU stays ($\beta \geq 1.2$), normalizing marginal benefit by expected bed-days (CMU-MGS) saves +1.42 to +4.99 additional expected lives over MGS ($p_{\text{Holm}} < 0.001$) by accelerating bed turnover.

(3) Robustness to Reneging Mortality Payoffs and Clinical Noise: When queue abandonment entails partial ($\theta_{\text{ren}} = 0.5$) or total ($\theta_{\text{ren}} = 0.0$) pre-ICU mortality, policies with high abandonment (MSF) collapse, whereas CMU-MGS remains the top-performing policy across all payoff regimes, expanding its advantage over greedy MGS to +5.41 lives at $\theta_{\text{ren}} = 0.5$. Similarly, under simultaneous Gaussian estimation noise ($\sigma_{\text{noise}} = 0.20$) across all clinical and LOS inputs, all five clinical policies maintain a highly significant advantage of +10.41 to +12.82 expected lives over FCFS ($p_{\text{Holm}} < 0.001$).

(4) Targeted Benefit of Partial Bed Pooling: Hierarchical overflow sharing between adult general and coronary care units provides substantial survival gains at balanced congestion ($\rho = 1.00$, saving +4.14 to +4.55 expected lives, $p_{\text{Holm}} < 0.001$), whereas its benefit vanishes under extreme overload ($\rho = 1.60$) once both units saturate simultaneously.

Data and Code Availability

All simulation source code, unit tests, DATASUS/CNES extraction files, configuration scripts, raw replication outputs (runs.csv, offline_ub.csv, and sensitivity grids), and plotting scripts are openly available in the project's public repository at https://github.com/MODAL-UFMA/alocacao-uti-web and archived with a permanent release tag (v3.0-cor-r3) and Zenodo metadata (.zenodo.json) for long-term reproducibility.

CRediT Authorship Contribution Statement

Carlos Eduardo Veras Gomes: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft. João Augusto Ferreira Nunes de Carvalho: Methodology, Software, Data curation, Visualization, Writing – original draft. Francisco Glaubos Nunes Clímaco: Conceptualization, Methodology, Formal analysis, Supervision, Project administration, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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